

Problem 8.13. Define $A_{\text{LBA}} = \{\langle M, w \rangle \mid M \text{ is an LBA that accepts input } w\}$. Show that A_{LBA} is PSPACE-complete.

Proof. To show that A_{LBA} is PSPACE-complete, we must show that it is in PSPACE and that all PSPACE problems are polynomial time reducible to it. The algorithm given in Theorem 5 decides A_{LBA} ¹. This algorithm runs in polynomial space and so $A_{\text{LBA}} \in \text{PSPACE}$.

To prove that A_{LBA} is PSPACE-hard, we give a polynomial time reduction from GG to A_{LBA} . This reduction converts a generalized geography graph G to an LBA L and a string w so that Player I has a winning strategy for the game played on G , iff L accepts w . The reduction F from GG to A_{LBA} uses a modified version L of algorithm M given in Theorem 8.14. The algorithm L takes input $\langle G, b, t \rangle$, where G is a directed graph, b is a node of G and t is a string. The algorithm L operates exactly as algorithm M , but only uses space occupied by t as its work memory. Hence, L is an LBA.

$F =$ “On input $\langle G, b \rangle$, where G is a directed graph, b is a node of G :

1. Compute the amount of space n required by M on input $\langle G, b \rangle$ ².
2. Construct L as described in proof idea above.
3. Let $w = \langle G, b, t \rangle$, where t is a string of length n .
4. Output $\langle L, w \rangle$.”

□

¹An LBA operates in exactly n space. This solution applies to both DLBA and NLBA due to Savitch’s Theorem.

²Space complexity analysis of algorithm M is given on page 346.